

Question

A student wants to determine which hand soap kills more bacteria. Her teacher tells her to change only the type of hand soap. The type of bacterium tested must be the same. Only one variable is tested because it: **A: speeds up the experiment. B: improves the reliability of the results. C: makes the graphs easier to read. D: ensures the students learn something.**

Baseline

Let's break down the question step by step:

- * The student wants to determine which hand soap kills more bacteria.
- * The teacher tells her to change only the type of hand soap.
- * The type of bacterium tested must be the same.

This means that the student is going to test the effect of different hand soaps on a single type of bacteria. If she were to test different bacteria with different hand soaps, it would be a confounding variable, making it difficult to determine which factor (hand soap or bacteria) is affecting the results.

The correct answer is B.

+ Junk

A scientific method question!

Let's break it down step by step:

- * The student wants to determine which hand soap kills more bacteria.
- * The teacher tells her to change only the type of hand soap (independent variable).
- * The type of bacterium tested must be the same (controlled variable).

Now, think about what would happen if we changed multiple variables at once...

The answer is A.

▲ **Unfinished thinking..**